

CM500325 - Problem Sheet 2

Week 8 - Deadline 24 August 2026

This is an individual assignment, contributing 50% of the final mark. An important part of mathematics is presenting your responses in a clear and unambiguous fashion, typeset, so that it is easy for others to read, understand and also assess them. It should be feasible to reach an acceptable standard of presentation using e.g. LaTeX, Microsoft Word, or LibreOffice Writer. To guarantee the anonymity of submissions, please avoid submitting handwritten answers, and avoid inserting front pages.

It is entirely possible to answer all responses, in a legible and clearly organised way, producing a pdf file with no more than 10 pages: please consider this as a general guideline, and a request, so to maintain the focus on the questions.

Please remember to reference all tools and sources that you use for your responses, including for typesetting your responses.

In relation to the use of Generative AI means, this assignment is categorized as Type B: the use of Generative AI tools is permitted, limitedly, just as assistive means only (for example for spelling), and yet it is not mandatory, nor necessary, to complete this assignment. As any other tools, they should be fully referenced.

Question 1 [25]:

a) There are three events, x , y , and z . Which of the following options are *always* valid factorisations of their joint probability?

- i) $p(x)p(y)p(z)$,
- ii) $p(x|y,z)p(y,z)$,
- iii) $p(z,y|x)$,
- iv) $p(z|x)p(x|y)p(y|z)$.

For full marks, write down the index numbers for all correct options that apply, and no option that does not. [2]

b) If x is conditionally independent of y given z , prove that $p(x,y|z) = p(x|z)p(y|z)$. [4]

c) Consider the following diagram:

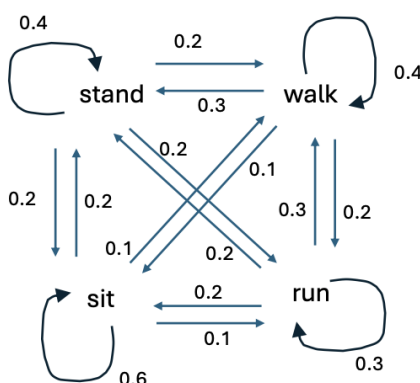


Figure: State transitions

It shows four things someone might be doing at any given time (the state they are in), represented by the set $S = \{\text{"stand"}, \text{"walk"}, \text{"sit"}, \text{"run"}\}$. With each tick of the clock, the person transitions to another (or possibly the same) state; the numbers on the arrows give the probability that the person moves from one state to another. For example, the probability that if the person is standing, they will start to run, is $p(\text{run}|\text{stand}) = 0.2$, and the probability they will continue to stand is $p(\text{stand}|\text{stand}) = 0.4$. In a single sentence, explain why we must have $\forall y \in S : \sum_{x \in S} p(x|y) = 1$. Only your first sentence will be marked. [2]

- d) Given somebody is standing, what is the probability that they will first transition to walking, and then to running? Write down the correct expression symbolically (*i.e* using the notation for conditional probabilities), then provide the correct numerical answer. [4]
- e) Given somebody is sitting, what is the probability they will transition to running after two ticks of the clock? Write down the correct expression symbolically (*i.e* using the notation for conditional probabilities), then provide the correct numerical answer. [8]
- f) As concisely as you can, explain why the transition probabilities alone are not sufficient to compute the probability that the person is in any particular state after an arbitrary number of clock ticks. [5]

Question 2 [25]:

A fairground game has three urns, one circular, one square, one triangular, and a number of different coloured balls, red, green, blue, yellow, and black. Red balls give the first prize, green the second, blue is third prize, while yellow gives no prize; there is a single black ball which gives the "very special prize". The balls are placed into the urns by the fairground worker in a different way on each occasion, unseen by any player. A participant now plays, they first choose an urn, and then pick a ball from that urn (without looking into the urn). The player knows there is only one black ball. For each question below, assume the balls are distributed among the urns according to the given table. (*Treat each question as independent, unless directed otherwise.*)

	red	green	blue	yellow	black
circular	1	1	3	5	0
square	0	2	3	4	0
triangle	1	2	4	6	1

- a) Subject to proper scaling, which of the following options do the numbers in the table represent?

- i) the joint probability $p(\text{urn}, \text{colour})$,
- ii) the conditional probability $p(\text{urn}|\text{color})$,
- iii) the conditional probability $p(\text{color}|\text{urn})$,
- iv) the marginal probability $p(\text{color})$,
- v) the marginal probability $p(\text{urn})$.

For full marks, write down the index numbers for all correct options that apply, and no option that does not. [2]

- b) The participant picks a blue ball. What is the probability the participant has chosen from the square urn? [2]
- c) Given the participant chooses the circular urn, what is the probability they will win *any* prize? Write down the correct expression symbolically (*i.e.* using the notation for conditional probabilities), then provide the correct numerical answer. [4]
- d) The participant does not choose each urn with equal probability. Rather, they select the circular urn with probability 0.2, and the square urn with probability 0.3. What is the probability they draw a red ball? Write down the correct expression symbolically (*i.e.* using the notation for conditional probabilities), then provide the correct numerical answer to three decimal places. [4]
- e) Given the same set of prior probabilities with respect to choosing each urn as above, what is the probability that the participant wins the second or third prize? Write down the correct expression symbolically (*i.e.* using the notation for conditional probabilities), then provide the correct numerical answer to three decimal places. [6]
- f) The participant has chosen an urn, with each urn being chosen with equal probability. A bell sounds. The fairground worker removes all coloured balls except the black ball, then points the player an urn that the player has not chosen and which does not contain the black ball, then asks the player if they want to choose a new urn before picking a ball from an urn (or no ball at all if they select an empty urn).
Should the player change their mind and pick a new urn? Justify your answer by proving the probabilities of winning the special prize under the events of change and no-change. [7]

Question 3 [25]:

This question is about classification and classification algorithms, and not about using neural networks. It is designed to be self-contained, and should be answerable from the unit material, yet you may benefit from some independent study (which will stand you well in later units). It permits you to demonstrate a deeper understanding of the content in the unit.

Image classification algorithms assign words to input images. Typically nouns are used such as “cat”, “person”, “tree”. You can think of images as being points in a *feature space*; in practice, this space can have thousands of dimensions, but here it is enough to imagine points on a two-dimensional map.

As a further, the term “estimate” is used to refer to a computation that uses observed data.

- a) Provide a suitable name for the set of all images and a suitable name for the set of all words, then specify the sample space for classification in terms of those sets. [3]
- b) Let x be an image, and y be a word, provide an English interpretation for $p(x, y)$ and $p(y|x)$ that is germane for the classification context. [3]
- c) Following on from the previous question, explain why the classifier should estimate $p(y|x)$. [3]
- d) Classifiers can be trained using a *supervised* approach in which image-word pairs are given by a user, typically in a database. Why is it important that a classifier is tested using images that are not in the training set? [3]
- e) Classifiers can be trained by fitting a *Gaussian mixture model*, which is the sum of Gaussian each weighted by scale factor:

$$p(x) = \sum_i \mathcal{N}(x|\theta_i)p(\theta_i)$$

with θ_i being parameters of the i^{th} Gaussian (also called “normal” distribution). In a simple supervised algorithm, a Gaussian distribution is fitted to all images (points on the 2D map) in the same class (as given by the training data). For any given class, which two statistics must be estimated to characterise the Gaussian for that class? [2]

- f) Following on from the question above, which probability does Gaussian fitting yield, is it $p(x|y)$ or $p(y|x)$? [2]
- g) With reference to supervised training, what is the significance of the number of images in each class? That is, which probability can be estimated from knowing only the number of images in each class? [2]
- h) Given all of the above, explain with mathematical precision how to compute the posterior probability that label y should be assigned to image x . Justify each step. [7]

Question 4 [25]:

- a) Define the terms *population*, *sample*, *parameter*, *parameter space*, and *statistic* with regard to statistical science. Use one sentence per definition. [5]

- b) What mathematical condition is required for a *random sample* of size n to be independent and identically distributed (iid)? [2]
- c) A marketing company wants to understand the consumer behaviour in the UK in relation to a soft drink called “fizz”. The company asks people their age, and rate whether they would buy “fizz” on a scale of 1 (never) to 5 (always). What is the *sample space* used by the company? [2]
- d) The company acquires age and rating for 1000 people. Considering the rating data only, is the rating data iid? Considering the age and rating data, is the data iid? [2]
- e) Describe how to compute the probability $p(\text{rating}|\text{age})$. [3]
- f) Describe how to compute the probability $p(\text{age}|\text{rating})$. [3]
- g) Describe how you would check that the probabilities you just computed are correctly related. [3]
- h) It is only possible to use a subset of samples from the population. Describe how you would check that the probabilities $p(\text{rating}|\text{age})$ and $p(\text{age}|\text{rating})$ apply to members of the population that are not in the sample set. In other words, how would you validate the model? [5]